Applied Econometrics: Session 3

Stats Review: The Most Dangerous Equation

PhD Intensive Course

Lecture 3

April 29, 2026

Why This Lecture Matters

Lectures 1–2 focused on causal meaning: potential outcomes, experiments, selection bias, and regression as a CEF approximation.

Today we add the missing statistical layer:

Core question

Once we have an estimate, how much should we trust its numerical value?

- A correct design can still produce a noisy estimate.
- Small samples generate surprisingly extreme averages.
- Standard errors, confidence intervals, tests, and p-values are tools for describing that noise.

Roadmap for Today

1. Sampling Variation
2. Standard Errors
3. Confidence Intervals
4. Hypothesis Tests and P-values
5. Uncertainty and Causal Estimates

The Three Objects Students Must Separate

Object	Meaning	Example
Estimand	Target in the population	$\mathrm{ATE} = \mathbb{E}[Y_1 - Y_0]$
Estimator	Rule applied to data	$\bar{Y}_1 - \bar{Y}_0$
Estimate	Number from this sample	4.2 test points

Beginner mistake

Treating the estimate as if it were the estimand. The estimate is one noisy draw from a sampling process.

One Sample Is Not the Whole Story

Imagine a population of students with a true average score $\mu = 70$.

- Draw one classroom of 25 students: the sample average might be 66.
- Draw another classroom of 25 students: the sample average might be 73.
- Neither classroom average is "the truth"; each is a noisy estimate.

Sampling variation

If we repeated the same sampling procedure many times, the estimate would vary from sample to sample.

The School-Ranking Trap

Suppose we rank schools by average test score.

- Some schools have 20 tested students.
- Others have 500 tested students.
- Even if all schools are equally good, the small schools will have much noisier averages.

Policy danger

The schools at the top of the ranking may look special partly because they had few students and a lucky draw.

Look at Both Tails

The mistake is clearer when we inspect both ends of the ranking.

- Small schools are overrepresented among the top performers.
- But small schools are also overrepresented among the bottom performers.
- This is a variance pattern, not evidence that small schools are automatically better or worse.

Diagnostic

When a small group appears often among the winners, check whether it also appears often among the losers.

The Most Dangerous Equation

For a sample mean,

$$\mathrm{SE}(\bar{Y}) = \frac{\sigma}{\sqrt{n}}.$$

- σ: individual-level standard deviation.
- n: number of observations used in the average.
- $\mathrm{SE}(\bar{Y})$: standard deviation of the estimate across repeated samples.

Translation

More data makes averages more stable, but precision improves with $\sqrt{n}$, not with n.

Appendix details

Why Precision Improves Slowly

If individual scores have $\sigma = 12$:

Sample size n	$\mathrm{SE}(\bar{Y}) = 12/\sqrt{n}$	Interpretation
25	2.40	very noisy average
100	1.20	half as noisy
400	0.60	half again
1600	0.30	half again

Rule: to cut the standard error in half, you need four times as many observations.

Standard Deviation vs Standard Error

Standard deviation	Standard error
Spread of individual outcomes	Spread of the estimator
Describes students, workers, firms	Describes $\bar{Y}$, $\hat{\beta}$, $\widehat{\mathrm{ATE}}$
Usually does not vanish with n	Shrinks as sample size grows
Answers: how different are units?	Answers: how precise is the estimate?

Practical warning

A dataset can have very dispersed outcomes and still estimate the mean precisely if n is large.

Estimators Are Random Variables

Before seeing data, $\bar{Y}$ is random because the sample is random.

$$\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i.$$

- Different samples produce different values of $\bar{Y}$.
- The sampling distribution describes those possible values.
- The standard error is the standard deviation of that sampling distribution.

Mental model

Think of the estimate in your paper as one point from a cloud of estimates that could have happened.

Appendix details

Bias and Variance Are Different Problems

$$\text{Estimation error} = \underbrace{\mathbb{E}[\hat{\theta}] - \theta}_{\text{bias}} + \underbrace{\hat{\theta} - \mathbb{E}[\hat{\theta}]}_{\text{sampling noise}} .$$

- Design assumptions target bias: selection, omitted variables, bad controls.
- Standard errors target sampling noise: how much $\hat{\theta}$ moves across samples.
- A tiny standard error does not make a biased comparison causal.
- A randomized experiment can be unbiased but still imprecise.

Python Demonstration for This Lecture

The file

`lectures/code/lecture-3-stats-review.py`

simulates two examples:

- school rankings where small schools appear in both tails because their averages are noisy;
- randomized treatment-control studies where the true treatment effect is fixed but estimates vary across samples.

Run:

```
python lectures/code/lecture-3-stats-review.py
```

Optional plot export:

```
python lectures/code/lecture-3-stats-review.py -save-plots
```

The Standard Error of a Mean

In practice σ is unknown, so we estimate it with the sample standard deviation:

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2}.$$

Then

$$\widehat{\mathrm{SE}}(\bar{Y}) = \frac{s}{\sqrt{n}}.$$

- Large s: outcomes are noisy, so the average is harder to estimate.
- Large n: averaging cancels out more individual noise.

Appendix details

A Small Numerical Example

Suppose average exam score in one class is $\bar{Y} = 72$, and $s = 12$.

Number of students	$\widehat{\mathrm{SE}}(\bar{Y})$	Rough 95% margin
25	$12/\sqrt{25} = 2.40$	$1.96 \times 2.40 = 4.70$
100	$12/\sqrt{100} = 1.20$	$1.96 \times 1.20 = 2.35$
400	$12/\sqrt{400} = 0.60$	$1.96 \times 0.60 = 1.18$

Same mean, same individual dispersion, very different precision.

Treatment-Control Difference in Means

In a randomized experiment with binary treatment D_i:

$$\widehat{\mathrm{ATE}} = \bar{Y}_1 - \bar{Y}_0,$$

where

$$\bar{Y}_1 = \frac{1}{n_1} \sum_{D_i=1} Y_i, \qquad \bar{Y}_0 = \frac{1}{n_0} \sum_{D_i=0} Y_i.$$

- Randomization gives the causal interpretation.
- The standard error tells us how noisy this causal estimate is.

Appendix details

Standard Error of a Difference

If treatment and control samples are independent,

$$\widehat{\mathrm{SE}}(\bar{Y}_1 - \bar{Y}_0) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}}.$$

- Each group mean is noisy.
- The variance of the difference adds the two group variances.
- The noisier or smaller group can dominate the uncertainty.

Causal translation

Even with random assignment, the estimated effect can look too large, too small, or even have the wrong sign in a small sample.

Appendix details

Unequal Group Sizes Waste Precision

Suppose $s_1 = s_0 = 12$ and total sample size is 200.

n_1 treated	n_0 control	$\widehat{\mathrm{SE}}(\bar{Y}_1 - \bar{Y}_0)$
100	100	1.70
50	150	1.96
20	180	2.83

Design lesson

For a fixed total sample and equal variances, balanced treatment-control groups are most precise.

What Standard Errors Do Not Do

Standard errors are essential, but they are not magic.

- They do not remove selection bias.
- They do not validate a bad control strategy.
- They do not prove the outcome was measured correctly.
- They do not solve spillovers, attrition, or noncompliance.

Good applied work needs both

Identification answers: is the estimate centered on the causal target? Inference answers: how uncertain is the estimate around its center?

The CLT as the Workhorse

Under common regularity conditions,

$$\frac{\bar{Y} - \mu}{\mathrm{SE}(\bar{Y})} \approx \mathcal{N}(0, 1).$$

- The sample mean is approximately normal in large samples.
- The approximation often works better for averages than for individual data.
- We use this to translate standard errors into intervals and tests.

Appendix details

A 95% Confidence Interval

For an estimate $\hat{\theta}$,

$$\hat{\theta} \pm 1.96\,\widehat{\mathrm{SE}}(\hat{\theta})$$

is the usual large-sample 95% confidence interval.

- Center: the estimate from this sample.
- Radius: about two standard errors.
- Wider interval: less precise estimate.
- Narrower interval: more precise estimate.

Appendix details

Treatment Effect Interval: Example

Suppose a randomized tutoring program produces:

$$\bar{Y}_1 - \bar{Y}_0 = 4.0, \qquad \widehat{\mathrm{SE}} = 1.5.$$

Then

$$95\%\ \mathrm{CI} = 4.0 \pm 1.96(1.5) = [1.06,\ 6.94].$$

Substantive reading

The data are compatible with a modest positive effect and a large positive effect, but not with exactly zero under the usual 95% rule.

What Does 95% Confidence Mean?

Frequentist confidence is about the procedure, not about a fixed interval.

- Imagine repeating the study many times.
- Each time, compute $\hat{\theta} \pm 1.96\,\widehat{\mathrm{SE}}$.
- About 95% of those intervals would cover the true parameter.

Careful language

After seeing one interval, the true parameter is fixed. The interval either covers it or it does not.

Appendix details

Common Confidence-Interval Mistakes

- Wrong: "There is a 95% probability that this exact interval contains the true parameter."
- Better: "This method covers the true parameter in about 95% of repeated samples."
- Wrong: "All values inside the interval are equally plausible."
- Better: "Values near the estimate are usually more compatible with the data than values near the edge."
- Wrong: "A wide interval means the design is not causal."
- Better: "A wide interval means the estimate is imprecise."

Do Not Compare Two Separate CIs

To test whether treatment and control means differ, use the CI for the difference:

$$(\bar{Y}_1 - \bar{Y}_0) \pm 1.96\sqrt{\frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}}.$$

- Separate CIs for $\bar{Y}_1$ and $\bar{Y}_0$ answer separate questions.
- Overlap of those separate CIs is not the right test for the treatment effect.
- The correct uncertainty is attached to the estimator you report.

Appendix details

Interval Width Is Information

Compare two estimates of the same causal effect:

Study	Estimate	95% CI
Small pilot	5.0	$[-3.0,\ 13.0]$
Large RCT	5.0	$[3.5,\ 6.5]$

Same point estimate, very different evidence.

- The pilot is compatible with harm, no effect, or large benefit.
- The large RCT pins down a positive effect much more tightly.

Hypothesis Testing: The Logic

A hypothesis test begins by assuming a benchmark is true.

$$H_0 : \theta = \theta_0.$$

Then ask:

Testing question

If H_0 were true, would an estimate this far from θ_0 be surprising?

- If very surprising, reject H_0.
- If not very surprising, do not reject H_0.
- "Do not reject" is not the same as "prove true."

Appendix details

The Test Statistic

For a large-sample normal test,

$$z = \frac{\hat{\theta} - \theta_0}{\widehat{\mathrm{SE}}(\hat{\theta})}.$$

- Numerator: distance from the null value.
- Denominator: typical sampling noise.
- A large absolute z means the estimate is far from the null relative to its noise.

95% two-sided rule

Reject H_0 at the 5% level if $|z| > 1.96$.

Appendix details

P-values

The p-value is:

$$\mathbb{P}(\text{test statistic at least this extreme} \mid H_0).$$

For a two-sided normal test:

$$p = 2\left[1 - \Phi\left(|z|\right)\right].$$

- Small p-value: this estimate would be unusual if H_0 were true.
- Large p-value: this estimate is not unusual under H_0.

Appendix details

The Most Common P-value Mistake

A p-value is not

$$\mathbb{P}(H_0 \mid \text{data}).$$

It is closer to

$$\mathbb{P}(\text{data as or more extreme} \mid H_0).$$

Plain English

A p-value asks how surprising the estimate is under the null. It does not tell us the probability that the null is true.

One-sided and Two-sided Tests

- Two-sided alternative:
$$H_1 : \theta \neq \theta_0.$$
Reject for estimates far above or far below the null.
- One-sided alternative:
$$H_1 : \theta > \theta_0 \quad \text{or} \quad H_1 : \theta < \theta_0.$$
Reject only in the pre-specified direction.

Applied convention

Use two-sided tests unless the one-sided direction was justified before seeing the data.

Statistical Significance Is Not Economic Importance

Result	Estimate	Standard error
Huge dataset	0.03	0.01
Small study	4.00	2.50

- The first effect may be statistically significant but economically tiny.
- The second effect may be substantively large but statistically imprecise.
- Applied interpretation must discuss magnitude, uncertainty, and design.

Failure to Reject Is Not Evidence of Zero

Suppose $\hat{\theta} = 4.0$ and $\widehat{\mathrm{SE}} = 3.0$.

$$z = \frac{4.0}{3.0} = 1.33, \qquad 95\%\ \mathrm{CI} = [-1.88,\ 9.88].$$

- We cannot reject zero at the 5% level.
- But the interval also contains effects large enough to matter.
- The study may simply be underpowered.

Better sentence

"The estimate is too imprecise to distinguish zero from meaningful positive effects."

Power: Why Small Studies Miss Effects

Power is the probability of rejecting H_0 when a particular alternative is true.

- Small n gives large standard errors.
- Large standard errors make $z = (\hat{\theta} - \theta_0)/\widehat{\mathrm{SE}}$ smaller in absolute value.
- Low-power studies often produce unstable estimates and inconsistent conclusions.

Applied implication

A noisy null result is not strong evidence that the causal effect is zero.

Multiple Testing and Specification Search

If you test many outcomes, subgroups, or specifications:

- some small p-values can arise by chance;
- the most impressive estimate is often selected after looking at many estimates;
- standard errors for one chosen regression do not account for all the searching.

Practical discipline

Pre-specify main outcomes when possible, report the full family of results, and be honest about exploratory analyses.

Design Credibility vs Statistical Precision

Problem	Main question
Identification	Is the estimator centered on the causal estimand?
Inference	How dispersed is the estimator around that center?
Measurement	Is the outcome/treatment recorded correctly?
External validity	Does this population answer the policy question?

Course habit

Never use the word "causal" just because a p-value is small.

Randomization Removes Selection, Not Luck

In an RCT:

$$D_i \perp\!\!\!\perp (Y_{1i}, Y_{0i})$$

gives unbiased comparison of treatment and control groups.

But in a finite sample:

$$\bar{Y}_1 - \bar{Y}_0$$

still varies across random assignments and random samples.

Lesson

Randomization solves the identification problem in expectation. Standard errors describe the remaining finite-sample uncertainty.

Appendix details

Observational Causal Designs Need More Care

In observational studies, uncertainty often has additional structure:

- heteroskedasticity: outcome variance differs across units;
- clustering: units in the same school, firm, village, or year share shocks;
- serial correlation: repeated observations over time are not independent;
- generated regressors: first-stage estimates add uncertainty.

Practical rule

The standard error formula must match the design and sampling process.

What to Report in Applied Work

For every main causal estimate, report:

- point estimate and units;
- standard error;
- confidence interval;
- sample size and group sizes;
- p-value if testing is relevant;
- the identifying assumption that gives the estimate causal meaning.

Minimum standard

An effect estimate without uncertainty is not yet an empirical result.

Reading a Results Table

When you see a table, read it in this order:

1. What is the estimand? ATE, ATT, regression slope, reduced form?
2. What design or assumption identifies it?
3. What is the sign and magnitude?
4. What is the standard error or confidence interval?
5. Is the estimate precise enough to answer the economic question?

Stars are last, not first.

A Useful Writing Template

Template

We estimate that X increases/decreases Y by $\hat{\theta}$ units ($\widehat{\mathrm{SE}} = s$), with a 95% confidence interval of $[L, U]$. The identifying variation comes from ....

Then interpret:

- Is $\hat{\theta}$ economically large?
- Does $[L, U]$ rule out policy-relevant alternatives?
- What assumption would move the estimate away from the causal target?

Lecture Takeaways

1. Small samples naturally create extreme estimates.
2. The standard error of a mean is $\sigma/\sqrt{n}$.
3. Confidence intervals describe uncertainty around an estimate.
4. Hypothesis tests ask whether data are surprising under a null.
5. P-values are $P(\text{data as extreme} \mid H_0)$, not $P(H_0 \mid \text{data})$.
6. Causal interpretation requires identification; statistical precision is a separate requirement.

Sources and Further Reading

- Matheus Facure Alves, *Causal Inference for the Brave and True*, Chapter 03: Stats Review: The Most Dangerous Equation.
- Howard Wainer (2007), "The Most Dangerous Equation," *American Scientist*, 95(3), 249–256.
- Angrist and Pischke, *Mostly Harmless Econometrics*, review of experiments and standard errors.
- Angrist and Pischke, *Mastering 'Metrics*, discussion of experiments and regression evidence.

These slides paraphrase and extend the chapter for this course; the examples use self-contained simulations.

Appendix Map

- A1: Variance of the sample mean
- A2: Sampling distributions
- A3: Sample standard deviation and plug-in SE
- A4: CLT and confidence intervals
- A5: Two-sample standard error
- A6: Hypothesis tests
- A7: P-value calculation
- A8: Why overlapping CIs are not the test
- A9: Randomization variance

Appendix A1: Variance of the Sample Mean

Let $Y_1, \ldots, Y_n$ be independent draws with

$$\mathbb{E}[Y_i] = \mu, \qquad \mathrm{Var}(Y_i) = \sigma^2.$$

The sample mean is

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^{n} Y_i.$$

Then

$$\mathrm{Var}(\bar{Y}) = \mathrm{Var}\left(\frac{1}{n} \sum_{i=1}^{n} Y_i\right) = \frac{1}{n^2} \sum_{i=1}^{n} \mathrm{Var}(Y_i) = \frac{\sigma^2}{n}.$$

Therefore

$$\mathrm{SE}(\bar{Y}) = \sqrt{\mathrm{Var}(\bar{Y})} = \frac{\sigma}{\sqrt{n}}.$$

Back to main slide

Appendix A2: Sampling Distributions

The population distribution describes individual Y_i. The sampling distribution describes an estimator such as $\bar{Y}$.

If

$$Y_i \sim \mathcal{N}(\mu, \sigma^2),$$

then exactly

$$\bar{Y} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right).$$

For non-normal Y_i, the central limit theorem gives the approximation

$$\bar{Y} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

in large samples.

Back to main slide

Appendix A3: Sample Standard Deviation and Plug-in SE

The unknown population variance is

$$\sigma^2 = \mathbb{E}[(Y_i - \mu)^2].$$

In data, replace μ with $\bar{Y}$:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (Y_i - \bar{Y})^2.$$

The denominator is $n - 1$, not n, because estimating $\bar{Y}$ consumes one degree of freedom.

Plug this into de Moivre's formula:

$$\widehat{\mathrm{SE}}(\bar{Y}) = \sqrt{\frac{s^2}{n}} = \frac{s}{\sqrt{n}}.$$

Back to main slide

Appendix A4: CLT and Confidence Intervals

The CLT implies

$$Z = \frac{\bar{Y} - \mu}{\mathrm{SE}(\bar{Y})} \approx \mathcal{N}(0, 1).$$

For a standard normal variable,

$$\mathbb{P}(-1.96 \leq Z \leq 1.96) \approx 0.95.$$

Substitute Z:

$$\mathbb{P}\left(-1.96 \leq \frac{\bar{Y} - \mu}{\mathrm{SE}(\bar{Y})} \leq 1.96\right) \approx 0.95.$$

Rearranging gives

$$\mathbb{P}\left(\bar{Y} - 1.96\,\mathrm{SE}(\bar{Y}) \leq \mu \leq \bar{Y} + 1.96\,\mathrm{SE}(\bar{Y})\right) \approx 0.95.$$

Back to main slide

Appendix A5: Two-Sample Standard Error

Let treatment and control sample means be independent:

$$\bar{Y}_1 = \frac{1}{n_1} \sum_{D_i=1} Y_i, \qquad \bar{Y}_0 = \frac{1}{n_0} \sum_{D_i=0} Y_i.$$

The estimator is

$$\hat{\tau} = \bar{Y}_1 - \bar{Y}_0.$$

Using $\mathrm{Var}(A - B) = \mathrm{Var}(A) + \mathrm{Var}(B) - 2\mathrm{Cov}(A, B)$:

$$\mathrm{Var}(\hat{\tau}) = \mathrm{Var}(\bar{Y}_1) + \mathrm{Var}(\bar{Y}_0) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0},$$

where independence makes the covariance term zero. Estimate with

$$\widehat{\mathrm{SE}}(\hat{\tau}) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}}.$$

Back to main slide

Appendix A6: Hypothesis Tests

To test

$$H_0 : \theta = \theta_0,$$

start from the approximate distribution

$$\hat{\theta} \approx \mathcal{N}\left(\theta, \mathrm{SE}(\hat{\theta})^2\right).$$

Under the null,

$$\frac{\hat{\theta} - \theta_0}{\mathrm{SE}(\hat{\theta})} \approx \mathcal{N}(0, 1).$$

Replacing the unknown standard error with its estimate gives

$$z = \frac{\hat{\theta} - \theta_0}{\widehat{\mathrm{SE}}(\hat{\theta})}.$$

For a two-sided 5% test, reject when $|z| > 1.96$.

Back to main slide

Appendix A7: P-value Calculation

Let $Z \sim \mathcal{N}(0,1)$, and suppose the observed test statistic is z_{obs}.

For a two-sided test,

$$p = \mathbb{P}(|Z| \geq |z_{\text{obs}}|) = 2\mathbb{P}(Z \geq |z_{\text{obs}}|) = 2\left[1 - \Phi(|z_{\text{obs}}|)\right].$$

Example:

$$z_{\text{obs}} = 2.3 \quad \Rightarrow \quad p = 2[1 - \Phi(2.3)] \approx 0.021.$$

This is small because values beyond ± 2.3 are rare under a standard normal null distribution.

Back to main slide

Appendix A8: Why Overlapping CIs Are Not the Test

Suppose two independent means have estimates $\bar{Y}_1, \bar{Y}_0$ and standard errors $\mathrm{SE}_1, \mathrm{SE}_0$. Separate 95% intervals are

$$\bar{Y}_1 \pm 1.96\mathrm{SE}_1, \qquad \bar{Y}_0 \pm 1.96\mathrm{SE}_0.$$

But the treatment effect is

$$\hat{\tau} = \bar{Y}_1 - \bar{Y}_0,$$

whose standard error is

$$\mathrm{SE}(\hat{\tau}) = \sqrt{\mathrm{SE}_1^2 + \mathrm{SE}_0^2}.$$

The correct 95% interval is

$$\hat{\tau} \pm 1.96\sqrt{\mathrm{SE}_1^2 + \mathrm{SE}_0^2}.$$

This object can exclude zero even when the two separate mean intervals overlap.

Back to main slide

Appendix A9: Randomization Variance

With potential outcomes

$$Y_i = D_i Y_{1i} + (1 - D_i) Y_{0i},$$

the difference in means is

$$\hat{\tau} = \bar{Y}_{1,\text{treated}} - \bar{Y}_{0,\text{control}}.$$

Under random assignment, treatment status is independent of potential outcomes, so

$$\mathbb{E}[\hat{\tau}] = \mathbb{E}[Y_{1i} - Y_{0i}]$$

for the relevant experimental population.

But the realized treated and control groups are finite random partitions, so

$$\mathrm{Var}(\hat{\tau}) \approx \frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0}.$$

Randomization centers the estimator correctly; it does not make the variance zero.

Back to main slide